

Beginning of the Universe Adjusted Assessment

The big bang

According to the most commonly accepted theory among cosmologists, the universe began about 15 billion years ago with a 'big bang'.

The Einstein connection

The big bang theory would not make any sense at all if it were not for Albert Einstein's famous equation. How could matter be created from nothing? Well, the singularity before the big bang was not 'nothing'. It was a huge amount of energy (with no mass) concentrated into a tiny, tiny point.

Einstein proposed that energy could be changed into matter. His equation $E = mc^2$ describes the change.

E represents the amount of energy in joules.

m represents the mass in kilograms.

c is the speed of light in metres per second (300 000 000 m/s).

Einstein's equation also describes how matter can be changed into energy. That is what happens in nuclear power stations, nuclear weapons and stars.

WORKING WITH BILLIONS AND TRILLIONS

One billion is equal to one thousand million; that is, 1 000 000 000 or 10^9 .

One trillion is equal to one thousand billion; that is, 1 000 000 000 000 or 10^{12} .

So one billion trillion is 1 000 000 000 000 000 000 000 or 10^{21} .

When numbers get that large, there are too many zeros to count. It is much easier to use powers of ten notation, also known as scientific notation.

The steady state theory

According to the steady state theory, which was proposed in 1948, there was no beginning of the universe. It was always there. The galaxies are continually moving away from each other. In the extra space left between the galaxies, new stars and galaxies are created. These new stars and galaxies replace those that move away, so that the universe always looks the same.

The great debate

A huge debate between those who supported the steady state theory and those who supported the big bang theory raged from 1948 until 1965. During that period, the evidence supporting the big bang theory grew.

The red shift

The red shift provides evidence for an expanding universe. This evidence supports the big bang theory and causes problems for those supporting the steady state theory. A steady state universe could expand only if new stars and galaxies replaced those that moved away. There is no way to explain how these new stars and galaxies could be created from nothing. Apart from that, these young stars and galaxies have not been found by astronomers.

The elements

The amounts of hydrogen and helium in the universe support the big bang theory. According to the steady state theory, the only way that helium can be produced is by the nuclear reactions taking place in stars.

About 8.7 per cent of the atoms in the universe are helium. This is far more than could be produced by the stars alone. The percentage of helium atoms can, however, be explained by their creation as a result of the big bang.

The afterglow

When George Gamow and Ralph Alpher proposed their version of the big bang theory in 1948, they

HOW ABOUT THAT!

The big bang theory was first proposed in 1927 by Georges Lemaitre, a Catholic priest from Belgium. But it wasn't called the 'big bang theory' then. Ironically, the name 'big bang' was invented by Fred Hoyle, one of the developers of the steady state theory. He used the name to try to ridicule the cosmologists who proposed the big bang theory.

In 1929, Edwin Hubble's observations and law provided convincing evidence to support Lemaitre's proposal. In 1933, Lemaitre presented the details of his theory to an audience of scientists in California.

Albert Einstein, by then recognised as one of the greatest scientists of all time, was in the audience. At the end of Lemaitre's presentation Einstein stood, applauded and announced, 'That was the most beautiful and satisfactory explanation of creation that I have ever heard'.



calculated that the universe would now, about 15 billion years after creation, have a temperature of 2.7°C above **absolute zero**. That's -270°C . Anything with a temperature above absolute zero emits radiation. The nature of the radiation depends on the temperature. Gamow predicted that, because of its temperature, the universe would be emitting an 'afterglow' of radiation. This afterglow became known as cosmic microwave background radiation.

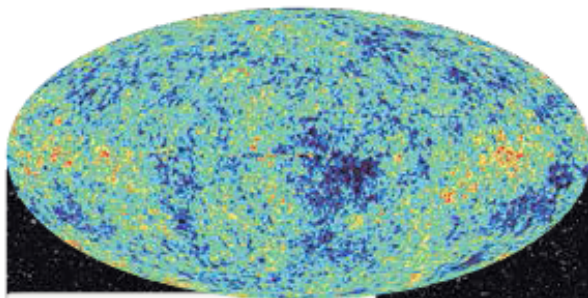
This radiation was discovered by accident in 1965. Engineers trying to track communications satellites picked up a consistent radio noise that they couldn't get rid of. The noise wasn't coming from anywhere on Earth, because it was coming from all directions. It was the cosmic microwave background radiation predicted by Gamow. Its discovery put an end to the steady state theory, leaving the big bang theory as the only theory supported by evidence currently available. Even Fred Hoyle, who had ridiculed the idea of a 'big bang', admitted that the evidence seemed to favour the big bang theory.

Mapping the universe

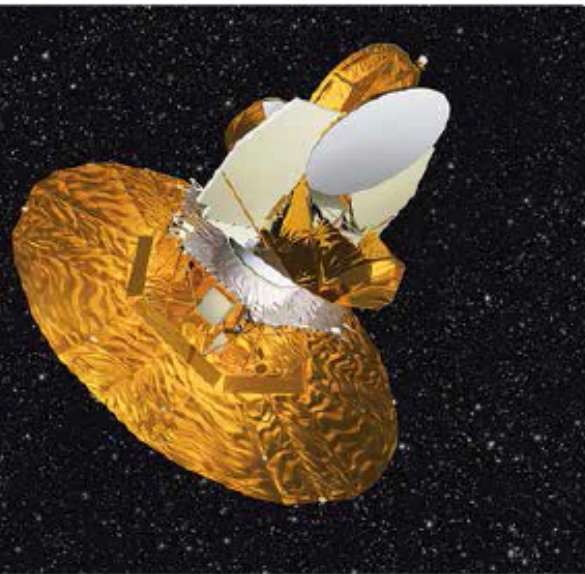
In 1989, a satellite named COBE (COsmic Background Explorer) was put into orbit around Earth to accurately measure the background radiation and temperature of the universe. COBE could detect variations as small as $0.000\,03^{\circ}\text{C}$. As predicted by Gamow, it detected an average temperature of -270°C .

In 2001, a probe called WMAP (Wilkinson Microwave Anisotropy Probe) was sent into orbit around Earth at a much greater distance to gather even more accurate data, detecting temperatures within a millionth of a degree. WMAP's first images were released by NASA in February 2003.

The computer-enhanced image of cosmic microwave background radiation shown below left was produced by the WMAP mission. The background radiation detected was released only 380 000 years after the big bang — the first radiation to escape. The image shows how the temperature varied across the universe as it was 380 000 years after the big bang. The blue parts of the map are the cooler regions. These regions were cool enough for atoms, and eventually galaxies, to form. The red parts are warmer regions. The map shows that galaxies are not evenly spread throughout the universe. They support the theory of an expanding universe that began with a big bang.



WMAP image of cosmic microwave background radiation



The Wilkinson Microwave Anisotropy Probe (WMAP). Its main mission was to gather evidence to help cosmologists find out how the universe began and predict what will happen in the future.

Will it ever end?

Will the expansion of the universe continue forever? If the universe does stop expanding, what will happen to it? There are several competing theories about the answers to these questions. One theory suggests that there is not enough mass in the universe for gravity to be able to pull it all back, so it will continue to expand forever. Other theories suggest that the universe will eventually end. According to these theories, the end will come when:

- the universe snaps back onto itself in a 'big crunch' (the **big crunch theory**). If this happens, the end result will be a single point — singularity. Some cosmologists believe that the big crunch will be followed by another big bang.

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Entropy: Is the end of the universe nearer than we thought?

Watch a video from the ABC's *Catalyst* program about the end of the universe.

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- the expansion of the universe continues and stars use up their fuel and burn out, causing planets to freeze (the **big chill theory**). The universe would then consist of scattered particles that never meet again.
- the universe rips itself apart violently as a result of expanding at an increasing speed (the **big rip theory**). According to this theory, the end of the universe will also be the end of time itself.

As you complete these questions, highlight the answers in the text above. Use Notability.

Questions

1. How old is the universe thought to be?

(1)

2. According to the Big Bang Theory, what was there at the time of the universe starting?

(1)

3. What does Einstein's equation ($E = mc^2$) have to do with the Big Bang Theory?

(1)

4. Who proposed the Big Bang Theory?

(1)

5. Which of the **two theories** about the "beginning" of the universe proposed that there was no beginning?

(1)

6. How did the Steady State Theory explain that the universe was expanding, yet the universe looked the same?

(1)

7. What piece of evidence discovered in 1965 put an end to the Steady State theory?

(1)

8. Give **one piece** of evidence that supported the Big Bang Theory.

(1)

9. Name and describe **three theories** about how the universe might end.

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(3)